

CoRoT-2b

R-1714

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_130830 · created 2026-08-02T15:52:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456534.8028 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.041
Transit depth (Rp/Rs)^2	3.6e-06 +/- 0.0015 [%]
Orbital Inclination (inc)	83.93 +/- 2.02
Airmass coefficient 1 (a1)	1.001 +/- 0.054
Airmass coefficient 2 (a2)	0.002 +/- 0.0076
Scatter in the residuals of the lightcurve fit is	5.4 %
Best Comparison Star	#2 - [105, 456]
Optimal Method	PSF photometry
Transit Duration (day)	0.057 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.0 ± 0.041 vs 0.1667 (deviation 4.07σ)
Duration fit vs published	0.057 d vs 0.095 d (deviation 1.65σ)
Mid-time O-C	-5.4 min
Depth SNR	0.0
Residual scatter	5.4 %

Light curve

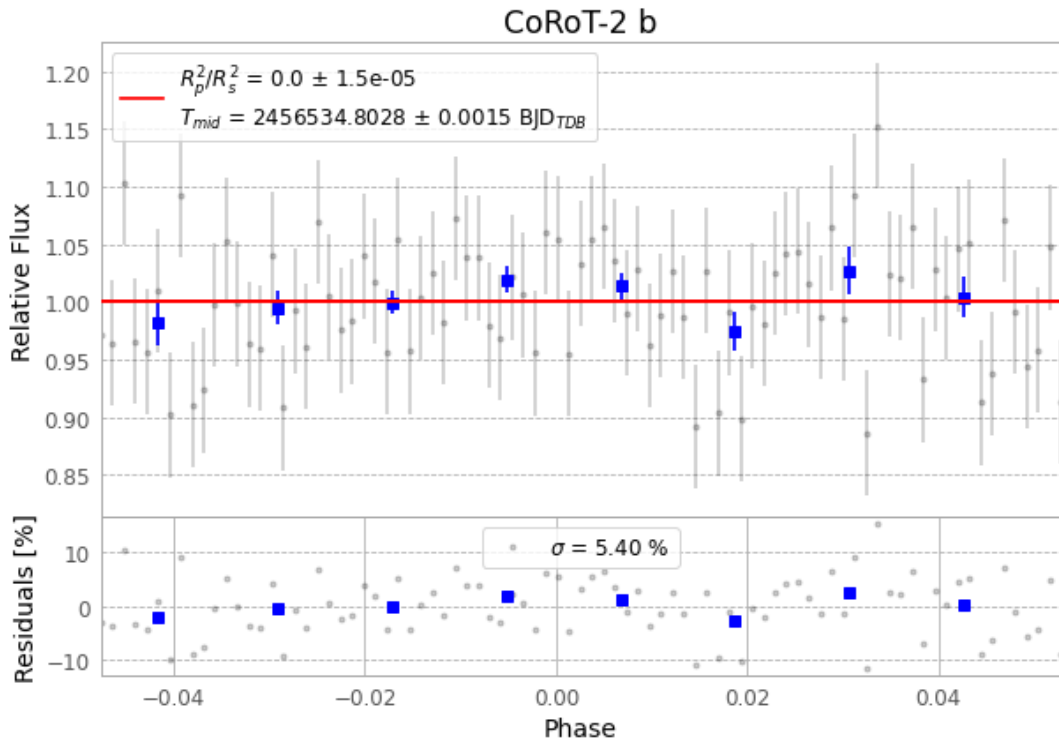


Figure 1 — FinalLightCurve_CoRoT-2 b_2013-08-29.png (SHA-256 8ed71d11d75deeab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 321], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 abb3e05c62971556...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), CoRoT-2130830050913.FITS → CoRoT-2130830092112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-130830.log (76671aa6ff9...), FinalLightCurve_CoRoT-2 b_2013-08-29.png (8ed71d11d75d...), FinalLightCurve_CoRoT-2 b_2013-08-29.csv (f2481303a650...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 15:52Z.